Practical - 12

Implementation of echo client server

AIM:

To implement echo client server using TCP/UDP sockets.

TCP code:

```
import socket
import threading

def start_server():
    server = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    server.bind(('127.0.0.1', 5000))
    server.listen(1)
    print("[SERVER] Listening on 127.0.0.1:5000")

    conn, addr = server.accept()
    print(f"[SERVER] Connected by {addr}")

    while True:
        data = conn.recv(1024).decode()
        if not data or data.lower() == 'quit':
            print("[SERVER] Connection closed by client")
            break
        print(f"[SERVER] Received: {data}")
        conn.send(f"Server received: {data}".encode())
    conn.close()
    server.close()

def start_client():
    client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
    client.connect(('127.0.0.1', 5000))
    print("[CLIENT] Connected to server")

    while True:
        msg = input("Enter message (or 'quit' to exit): ")
        client.send(msg.encode())
        if msg.lower() == 'quit':
            break
```

```
        response = client.recv(1024).decode()
        print(f"[SERVER RESPONSE] {response}")
    client.close()

if __name__ == "__main__":
    server_thread = threading.Thread(target=
        start_server, daemon=True)
    server_thread.start()

    start_client()
```

OUTPUT:

[SERVER] Listening on 127.0.0.1:5000 [CLIENT] Connected to server

[SERVER] Connected by ('127.0.0.1', 55659) Enter message (or 'quit' to exit): computer

[SERVER] Received: computer

[SERVER RESPONSE] Server received: computer

Enter message (or 'quit' to exit): Hi

[SERVER] Received: Hi

[SERVER RESPONSE] Server received: Hi

Enter message (or 'quit' to exit): quit

[SERVER] Connection closed by client.

UDP code 1-

```
import socket
import threading

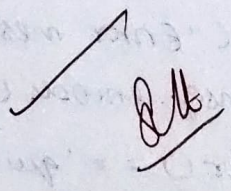
def udp_server():
    s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    s.bind(('127.0.0.1', 12000))
    print("[SERVER] Running on 127.0.0.1:12000")
    while True:
        data, addr = s.recvfrom(1024)
        msg = data.decode()
        if msg.lower() == 'quit':
            print(f"[SERVER] Client {addr} ended the
                  connection")
            break
        print(f"[SERVER] Received from {addr}: {msg}")
        s.sendto(data, addr)
    s.close()
    print("SERVER stopped")

def udp_client():
    c = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)
    server = ('127.0.0.1', 12000)
    while True:
        msg = input("Enter message (or 'quit' to stop): ")
        c.sendto(msg.encode(), server)
        if msg.lower() == 'quit':
            print("[CLIENT] Closing connection...")
            break
        data, _ = c.recvfrom(1024)
        print("Echo from server:", data.decode())
    c.close()
    print("[CLIENT] Stopped")
```

```
if __name__ == "__main__":
    threading.Thread(target=udp_server,
        daemon=True).start()
    server_thread.start()
    udp_client()
```

OUTPUT:

```
[SERVER] Running on 127.0.0.1:12000
Enter message (or 'quit' to stop): computer
[SERVER] Received from ('127.0.0.1', 55598): computer
Echo from server : computer
Enter message (or 'quit' to stop): quit
[SERVER] client ('127.0.0.1', 55598) ended the
connection
[CLIENT] closing connection...
[CLIENT] stopped [SERVER] stopped.
```



RESULT:

Thus the echo client-server communication was successfully implemented using TCP and UDP sockets, demonstrating end-to-end communication at transport layer